

# Class overview and logistic regression

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# Welcome to STA 712!

## Agenda:

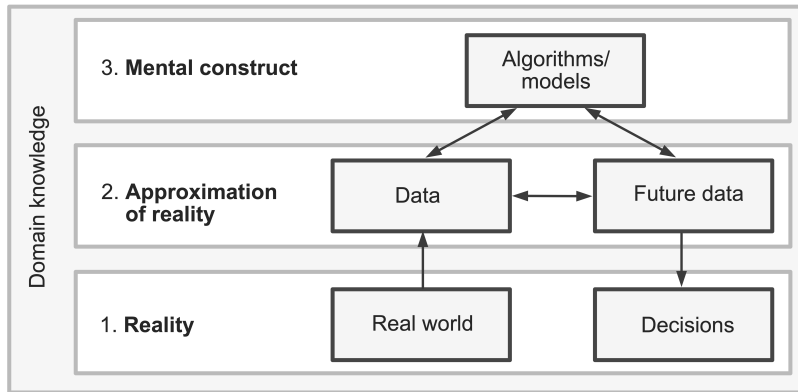
- ▶ Brief course overview and syllabus highlights
- ▶ Sketch plan for first couple weeks
- ▶ Logistic regression

# Learning goals

- ▶ **Choosing reasonable methods:** select and fit an appropriate model, and use that model to answer questions of interest
- ▶ **Statistical thinking:** curious about the data, skeptical about the limitations of our methods, and cautious about our conclusions
- ▶ **“What can I do differently?”:** we need to adapt to the data and the situation, not the other way around!
- ▶ **Theory and simulation** to characterize properties and assumptions of statistical models, and explore consequences of different modeling choices
- ▶ **Communicating** results in context

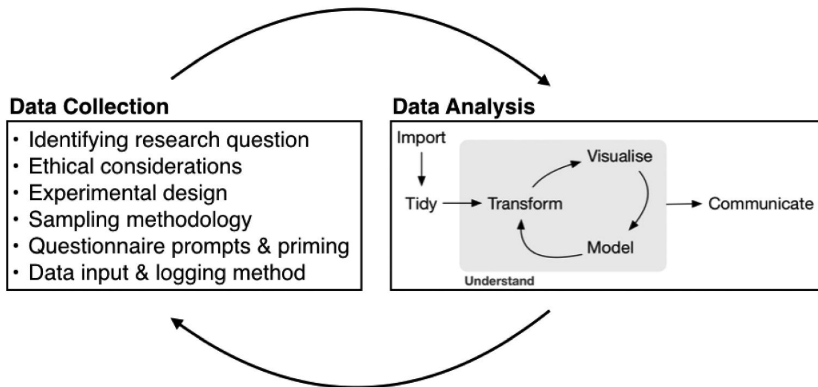
# The role of modeling

*Essentially, all models are wrong, but some are useful.* (Box and Draper, 1987)



(Image from *Veridical Data Science*, Yu and Barter)

# Components of (good) data analysis



Albert Y. Kim & Johanna Hardin (2021) "Playing the Whole Game": A Data Collection and Analysis Exercise With Google Calendar, *Journal of Statistics and Data Science Education*, 29:sup1, S51-S60, DOI:

10.1080/10691898.2020.1799728

# Overview of course content

- ▶ Core types of models: GLMs (and variants)
  - ▶ Logistic regression (binary responses), Poisson regression (count responses), linear regression (some continuous responses)
  - ▶ GLM theory
- ▶ Model extensions/generalizations
  - ▶ Overdispersion in count data: quasi-Poisson and negative binomial models
  - ▶ Excess 0s: zero-inflated models
  - ▶ Dependent data: mixed models (time permitting)
- ▶ Model assessment
  - ▶ Diagnostics for model assumptions
  - ▶ Assessing model predictions
- ▶ Inference (hypothesis tests and confidence intervals)
  - ▶ Wald and LRT
  - ▶ Parametric bootstrap (time permitting)

## Course structure

- ▶ Class participation and seminar attendance
- ▶ Assigned readings
- ▶ Regular HW assignments (due most weeks)
- ▶ Homework presentations (3)
- ▶ One midterm exam
- ▶ Project (throughout second half of semester)
- ▶ Final exam

# Plan for the first couple weeks

## Recap logistic regression

- ▶ Motivation and interpretation
- ▶ Estimation
- ▶ Inference

## Model assessment

- ▶ Assumptions and diagnostic plots
- ▶ Outliers and multicollinearity
- ▶ Assessing predictions

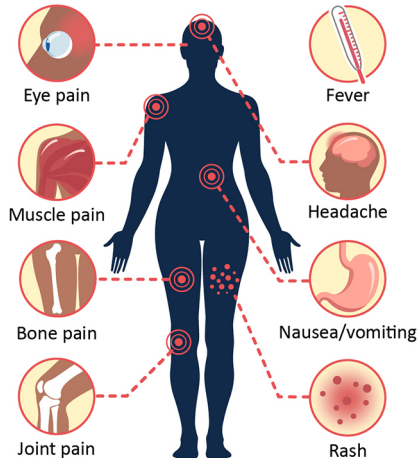


# Motivating example: Dengue fever

**Dengue fever:** a mosquito-borne viral disease affecting 400 million people a year

## Dengue Symptoms

Fever with any of the following



## Motivating example: Dengue data

**Data:** Data on 5720 Vietnamese children, admitted to the hospital with possible dengue fever. Variables include:

- ▶ *Sex*: patient's sex (female or male)
- ▶ *Age*: patient's age (in years)
- ▶ *WBC*: white blood cell count
- ▶ *PLT*: platelet count
- ▶ other diagnostic variables. . .
- ▶ *Dengue*: whether the patient has dengue (0 = no, 1 = yes)

**Research goal:** Predict dengue status using diagnostic measurements

**Question:** What type of model should we use, and why?

# Logistic regression model

The original authors chose the following model:

$$Dengue_i | Age_i, WBC_i, PLT_i \sim \text{Bernoulli}(p_i)$$

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 Age_i + \beta_2 WBC_i + \beta_3 PLT_i$$

**Question:** Why is there no noise term  $\varepsilon_i$  in the logistic regression model?

## Logistic regression model

```
m1 <- glm(Dengue ~ Age + WBC + PLT,  
          data = dengue, family = binomial)  
summary(m1)$coefficients
```

| ##             | Estimate     | Std. Error   | z value    | Pr      |
|----------------|--------------|--------------|------------|---------|
| ## (Intercept) | 1.252559319  | 0.1548038204 | 8.091269   | 5.90463 |
| ## Age         | 0.138318635  | 0.0099762568 | 13.864783  | 1.03538 |
| ## WBC         | -0.252329417 | 0.0135371014 | -18.639841 | 1.52694 |
| ## PLT         | -0.006027573 | 0.0006112861 | -9.860478  | 6.17549 |

**Question:** How would I interpret the estimated coefficient

$\hat{\beta}_1 = 0.1383$  for Age?

## Another example

Examining a health survey in the Netherlands, researchers saw a potential association between having a pet bird and developing lung cancer. They now want to conduct a follow-up study to further examine whether individuals who keep pet birds are at greater risk.

### Questions:

- ▶ How would you try to design this study?
- ▶ What data would you want to collect?
- ▶ What model would you want to use?

## Class activity

```
## (Intercept)          Bird      SexMale      SocEcLow
## -1.27063830  1.36259456 -0.56127270 -0.10544761 -0.03975
##      SmokeRate
##    0.02601689
```

Interpret the estimated coefficient 1.363 in context.

## Class activity

```
## (Intercept)          Bird      SexMale      SocEcLow
## -1.27063830   1.36259456 -0.56127270 -0.10544761 -0.03975
##      SmokeRate
##      0.02601689
```

The estimated coefficient on Age is negative. Why might that be the case? Is there anything you would change in the model?

## Class activity

```
## (Intercept)          Bird      SexMale      SocEcLow
## -1.27063830   1.36259456 -0.56127270 -0.10544761 -0.03975
##   SmokeRate
##   0.02601689
```

What is the predicted probability of lung cancer for a male individual who keeps a bird, of “high” socioeconomic status, aged 35, who has never smoked?



## Class activity

```
## (Intercept)          Bird      SexMale      SocEcLow
## -1.27063830   1.36259456 -0.56127270 -0.10544761 -0.03975
##   SmokeRate
##   0.02601689
```

$$\frac{\exp\{-1.271 + 1.363 - 0.561 - 0.040(35)\}}{1 + \exp\{-1.271 + 1.363 - 0.561 - 0.040(35)\}} = 0.134$$